

FINAL PROJECT

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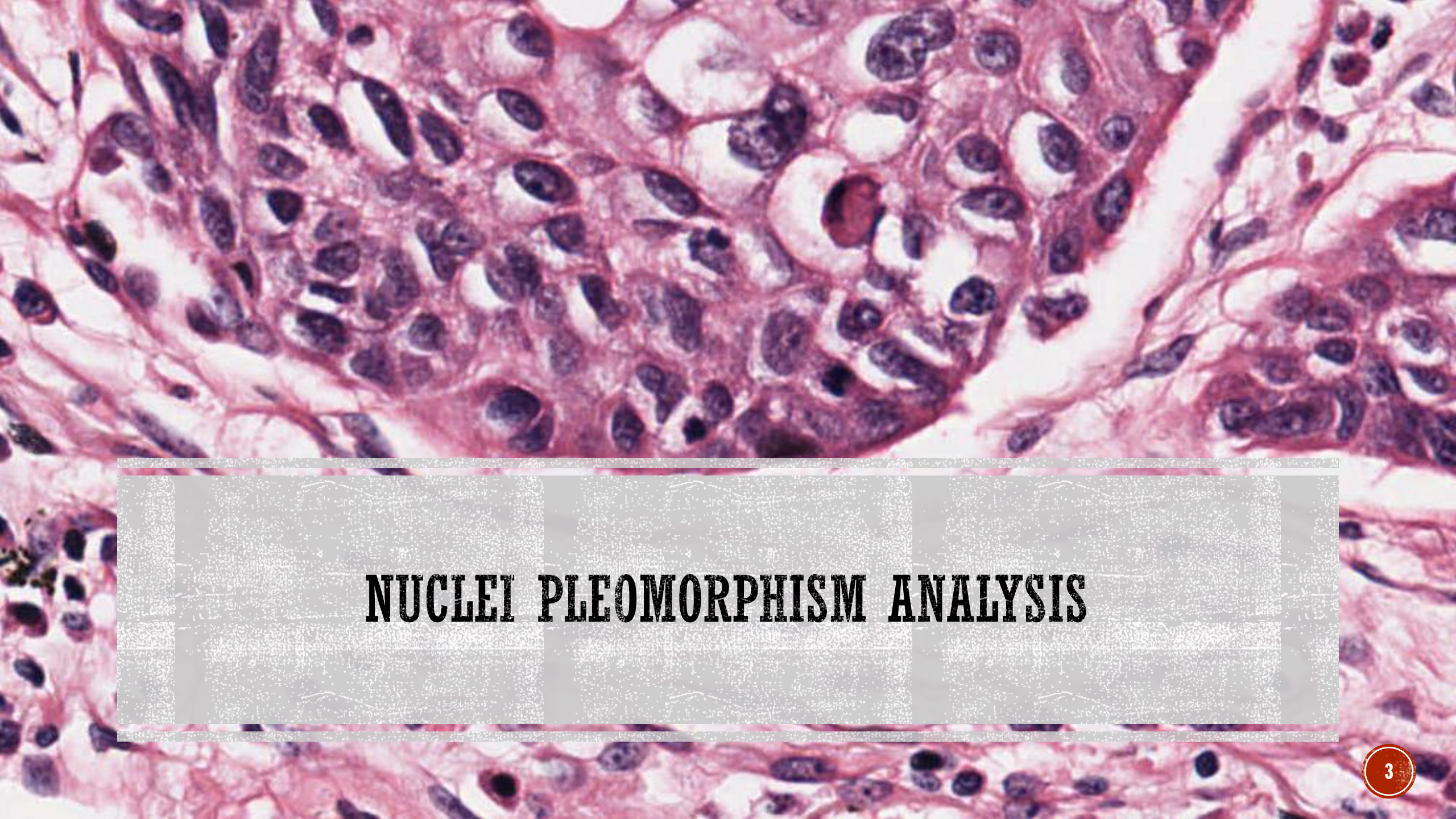


OUTLINE

- Histogram
- Image Enhancement
- Image Filtering
- Image Restoration
- Masking
- Morphological Filters
- Edge Detection
- Image Sharpening
- Corners
- Line and Circles Detection
- Distance Transform
- Watershed Segmentation

Sebelum ETS

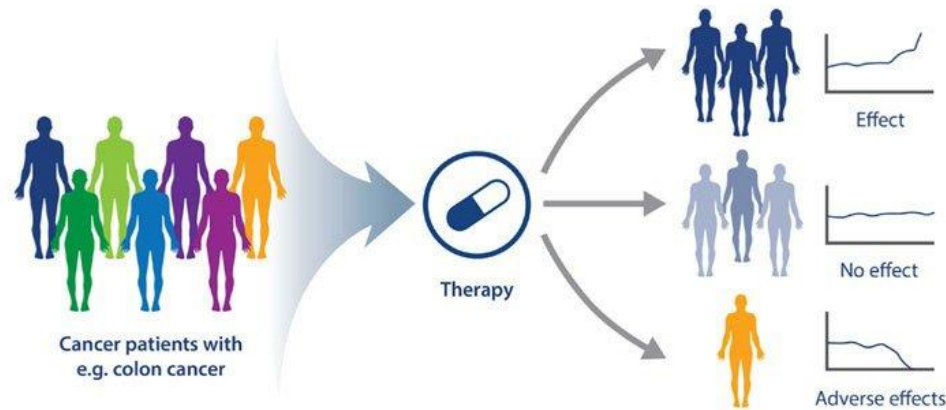
Setelah ETS

A histological slide showing a dense population of cells with prominent nuclear pleomorphism. The nuclei vary significantly in size and shape, with some being large and hyperchromatic, and others smaller and more uniform. The background is a pinkish-red stain, likely hematoxylin and eosin (H&E).

NUCLEI PLEOMORPHISM ANALYSIS

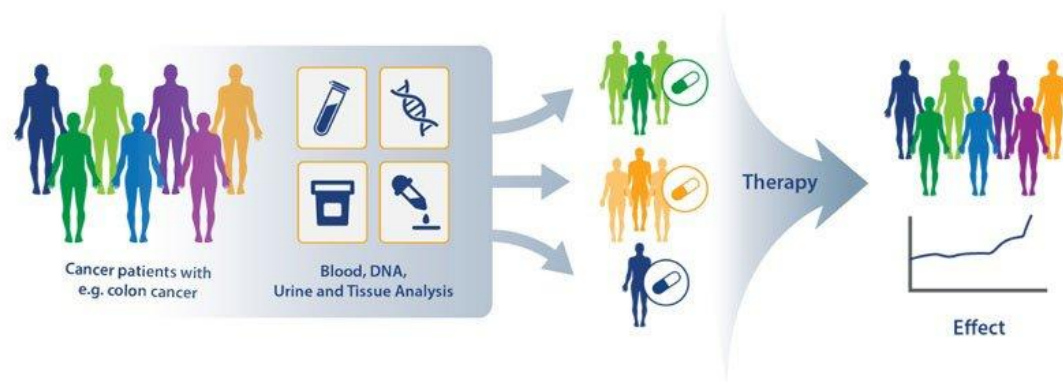
Traditional Medicine

One Treatment Fits All



Precision Medicine

More Personalized Diagnostics



PRECISION MEDICINE?

- **Traditional medicine:** “One-Size-Fits-All” or Population-Average model.
- Relies on clinical trials where a treatment is tested and if the majority show improvement → “standard of care”
- **Precision medicine:** “Data-Driven Individualization” model
- Instead of treating as an average member of a large group → treats as a member of specific sub-population

KEY STRUCTURAL DIFFERENCES

Characteristics	Traditional Medicine	Precision Medicine
Approach	Based on population average (One-size-fits-all).	Based on individual specific profiles (Customized).
Diagnostic Methods	Clinical symptoms and micro/macro visual observations	Integration of genetic, molecular data and quantitative image analysis
Therapeutic Impact	The risk of overtreatment or the drug not working is high	More effective therapy with minimal side effects

WHY THE SHIFT IS HAPPENING?

- The limitation of traditional medicine: “Hidden minority”
- Example: in many common drugs for depression or high blood pressure, a significant percentage of patients do not respond to the first drug prescribed.
- Precision medicine aims to eliminate this by using:
 - **Genomics:** Understanding how your genes affect drug metabolism.
 - **Digital Pathology:** Using deep learning to analyze tissue biopsies with a level of detail that identifies unique disease signatures.
 - **Wearable Data:** Monitoring real-time environmental and lifestyle factors.

NUCLEI PLEOMORPHISM?

- Definition: Abnormal variations in the size, shape, and color (texture intensity) of cell nuclei within a tissue.
- Normal Condition: **Cell nuclei are uniform**, neatly round/oval, and nearly the same size.
- Cancerous Condition (Pleomorphic): Cell nuclei **are irregularly shaped** (some are very oval or wrinkled), **vary drastically in size** (some are giant, some are small), and the **chromatin within them is very dark/coarse**.

NUCLEI PLEOMORPHISM?

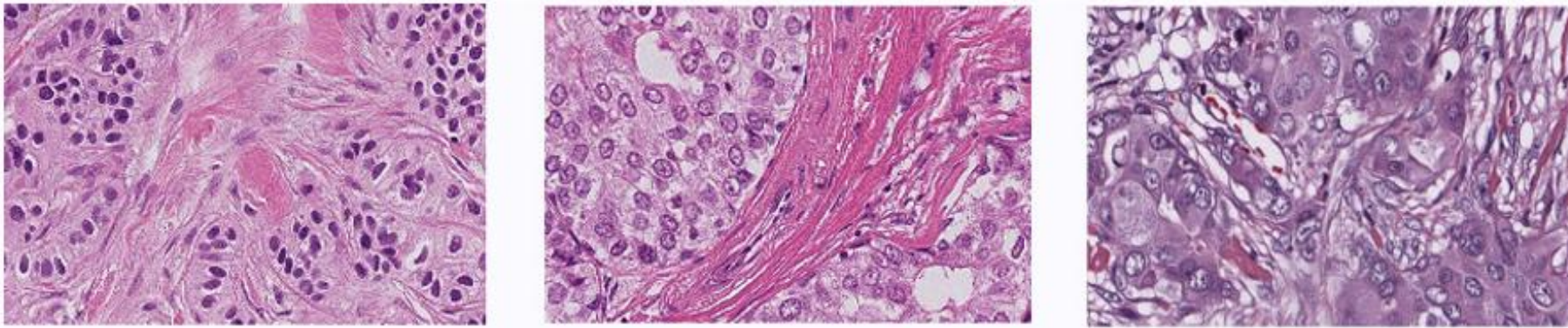
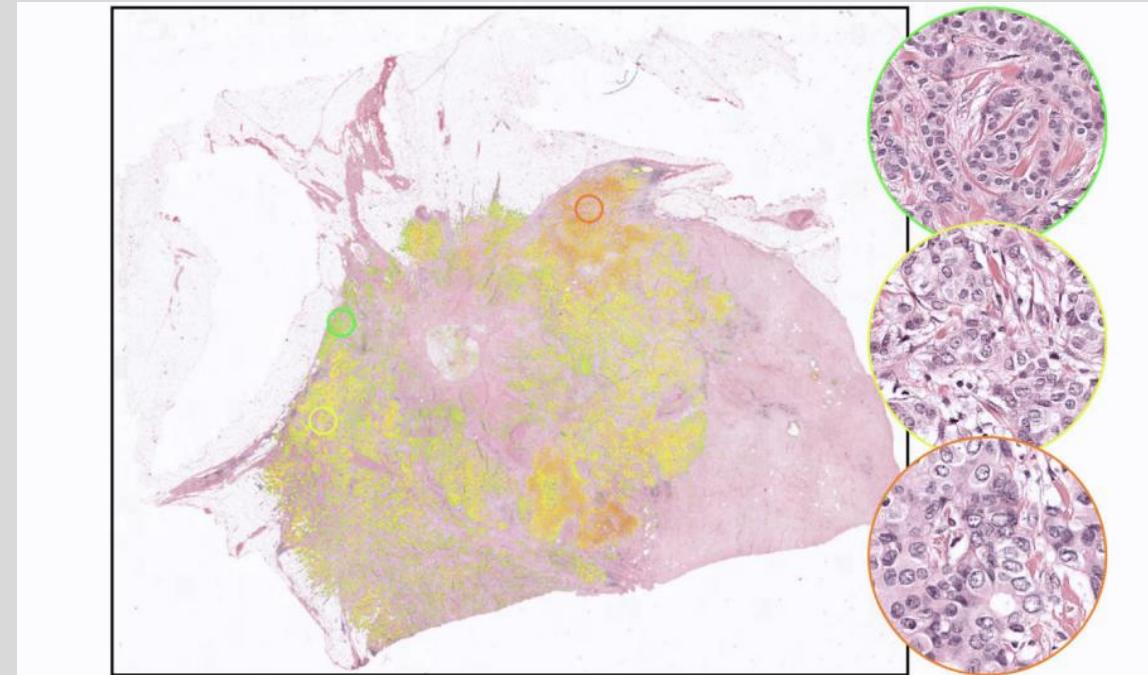
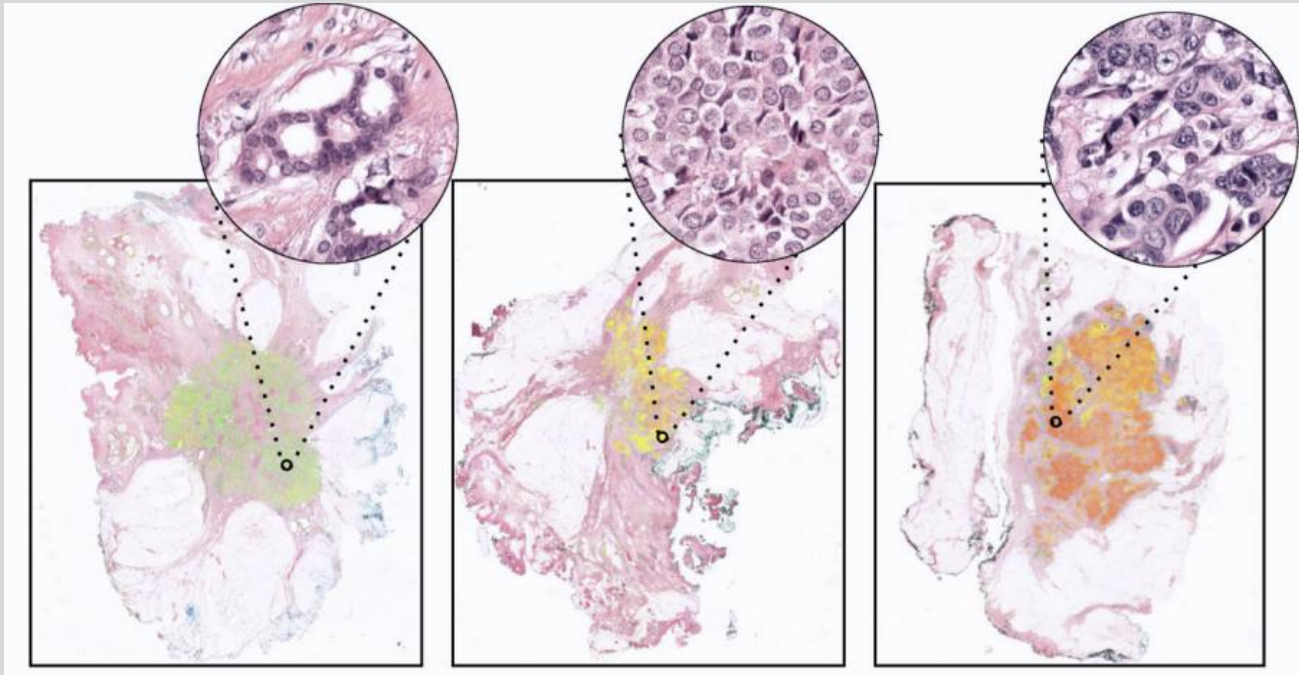


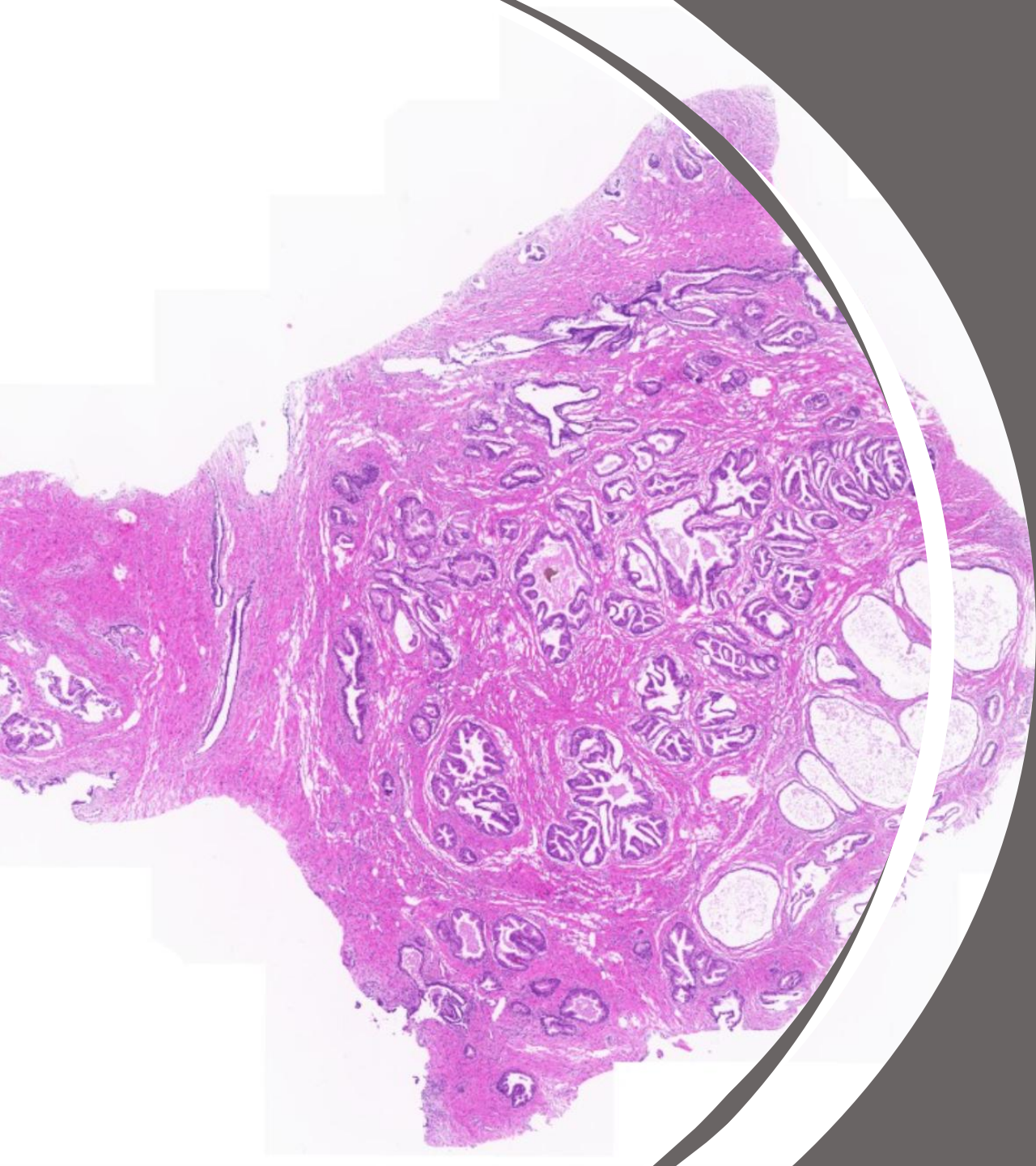
Fig. 1. Nuclear pleomorphism scores. **Left:** Nuclear atypia grade 1. **Center:**, grade 2, **Right:** grade 3

NUCLEI PLEOMORPHISM?



WHY IS IT IMPORTANT?

- Highly pleomorphic nuclei → often indicate genomic instability → linked to specific mutations and pathways → guide targeted therapies
- Precision medicine relies heavily on genomics (e.g., sequencing tumors), but morphology still plays a role.
- Nuclear pleomorphism often correlates with:
 - High mutation burden
 - Aggressive molecular subtypes
 - Alterations in genes like TP53, HER2, etc.



DIGITAL PATHOLOGY & PRECISION MEDICINE

In Precision Medicine, phenotype (cell shape) reflects genotype (DNA damage). With Digital Pathology, we can:



Extract invisible features.

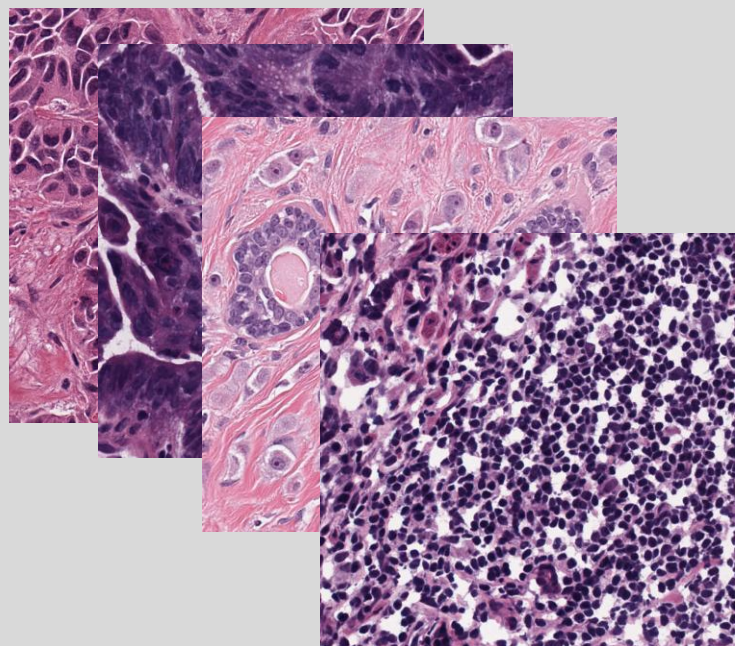


Standardize diagnostic criteria.



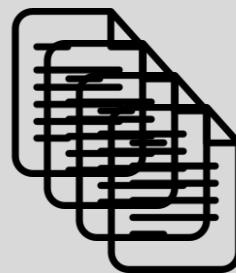
Predict molecular profiles non-invasively

DATASET: MONUSEG 2018

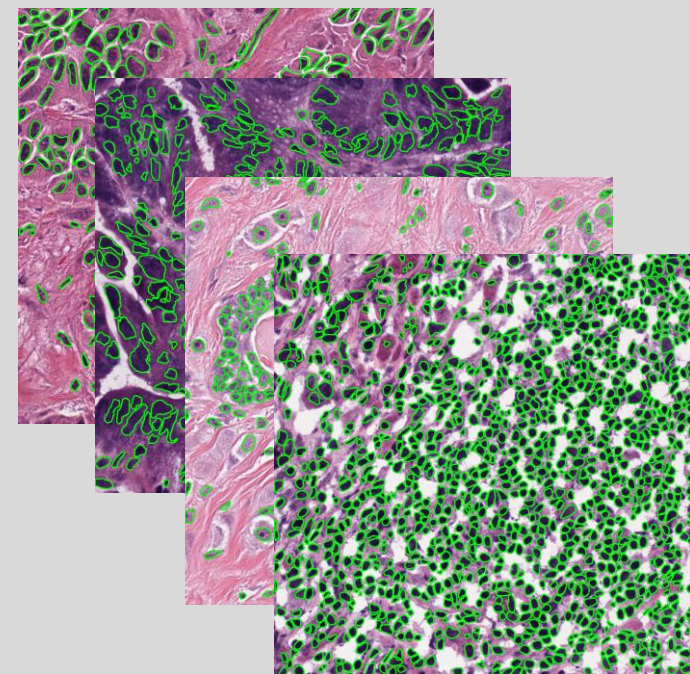


Tissues.tif

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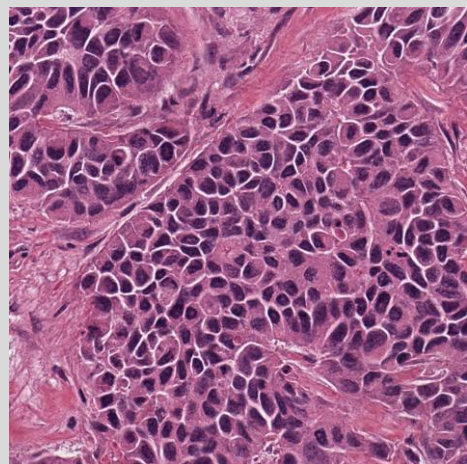


Ground Truth.xml



Overlay

FINAL PROJECT

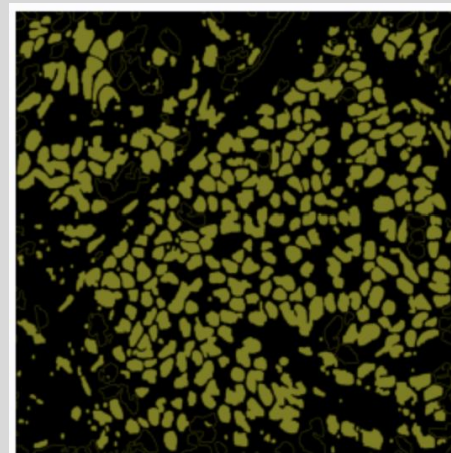


Original

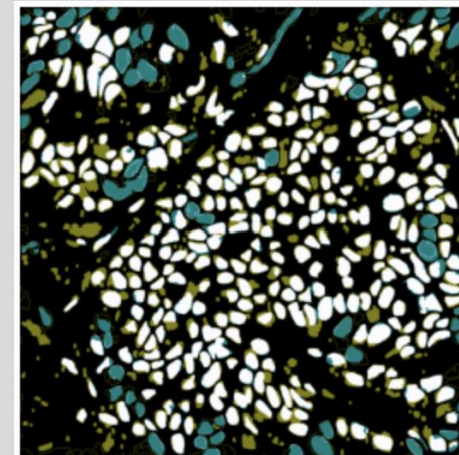
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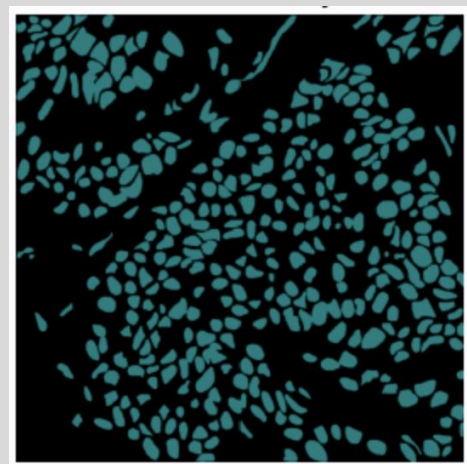
Algorithm



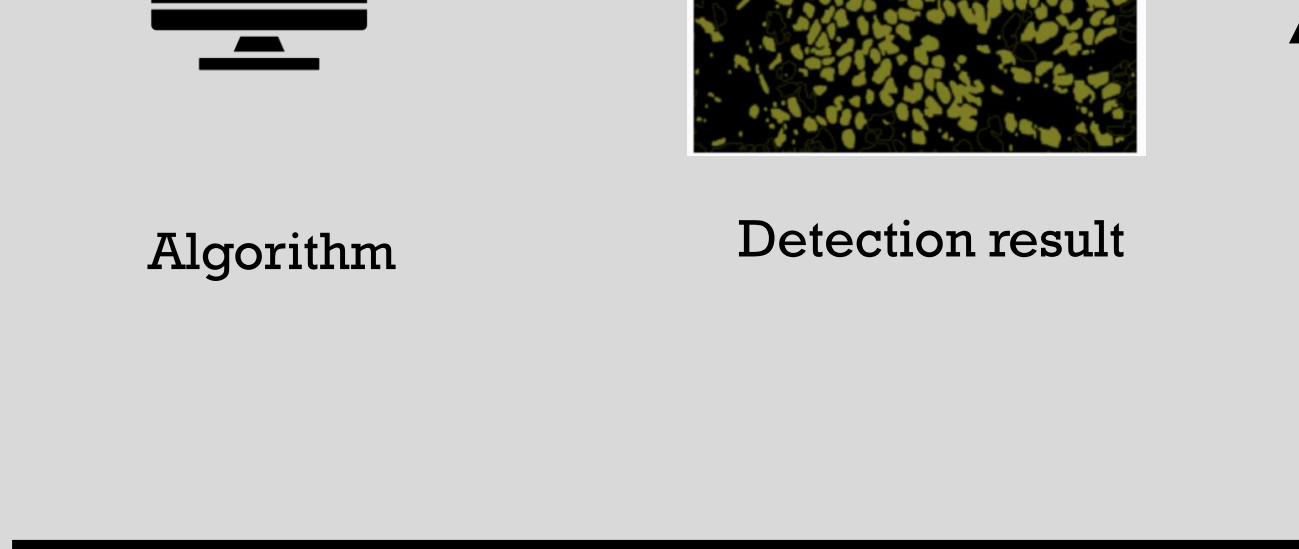
Detection result



Overlay



Ground truth



FINAL PROJECT

- Dari dataset MoNuSeg2018 yang telah di upload di MyITS Classroom, buatlah program untuk mendeteksi sel nucleus pada masing masing citra
- Desainlah sebuah system yang dapat melakukan segmentasi sel nucleus secara otomatis dengan menggunakan pengolahan citra dasar
- Lakukan analisis akurasi dengan menggunakan metrik Jaccard Index (IoU), Dice Similarity, dan Running Time
- Submit program anda dalam bentuk GUI dan laporan teknis (format.pdf) ke MyITS Classroom, kemudian masukkan hasil evaluasi (IoU, Dice Similarity dan Running Time) kedalam link: <https://its.id/FormLeaderboardPCM>
- Note: multiple submit in the form leaderboard are welcome, just give the version name (ex: Nabila_Ver1)
- **Deadline demo: 17 Juni 2026**

THANK YOU